

Architecture en couches du backend Spring Boot

HTTP layer — Controllers + Security + Swagger

SiteController

TerrainController

MembreController

MatchController

ReservationController

PaielementController

AuthController

@RestController · @RequestMapping · @Valid · @Tag · @Operation · HTTP verbs GET/POST/PUT/PATCH/DELETE



DTO layer — Request / Response / Mapper

SiteRequest/Response

TerrainRequest/Response

MembreRequest/Response

MatchRequest/Response

ReservationRequest/Response

PaielementResponse

LoginRequest/Response

Never expose JPA entities directly · @NotNull · @NotBlank · @Email · @Future · one Mapper per entity



Service layer — Business logic (interface + impl)

SiteService

TerrainService

MembreService

MatchService

ReservationService

PaielementService

AuthService

SOLID · Dependency Inversion · @Service · @Transactional · @RequiredArgsConstructor · @Slf4j · BookingDelay · PenaltyCheck · BalanceCheck



Repository layer — JPA interfaces

SiteRepository

TerrainRepository

MembreRepository

AdministrateurRepository

MatchRepository

ReservationRepository

PaielementRepository

PenaliteRepository

JourFermetureRepository

extends JpaRepository · Spring generates SQL from method names · findBy · existsBy · countBy



Entity layer — JPA entities + Enums

Site

Terrain

Membre

Administrateur

Match

Reservation

Paielement

Penalite

JourFermeture

BaseEntity

TypeMembre

TypeAdministrateur

TypeMatch

StatutMatch

StatutReservation

StatutPaielement

@Entity · @MappedSuperclass · @OneToMany · @ManyToOne · @OneToOne · @Enumerated(STRING) · FetchType.LAZY · CascadeType.ALL · @EnableJpaAuditing

Config layer

SecurityConfig

JwtConfig

JwtAuthenticationFilter

SwaggerConfig

CorsConfig

JWT · BCrypt · STATELESS · Bearer Auth · CORS localhost:4200

Infrastructure

MatchScheduler

DataInitializer

GlobalExceptionHandler

ErrorResponse

docker-compose.yml

@Scheduled cron · CommandLineRunner · @RestControllerAdvice · PostgreSQL 15 Docker

Test layer — Unit tests (Mockito + JUnit 5 + AssertJ)

MembreServiceTest — 23 tests

MatchServiceTest — 26 tests

ReservationServiceTest — 18 tests

PaielementServiceTest — 18 tests

@ExtendWith(MockitoExtension) · @Mock · @InjectMocks · @Nested · @DisplayName · @BeforeEach · assertThatThrownBy · verify · H2 in-memory for integration · 87 tests total